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ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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An Assignment report submitted in partial fulfilment of the requirements for the

CSA0315 - Data Structures

to the award of the degree of

BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE ENGINEERING

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Title

Design of a Collision-Resilient Hash-Based Caching System for High-Throughput Product Lookups in E-Commerce

1. Problem Understanding & Formulation

E-commerce platforms process a large number of product lookup requests. Searching products directly from the main database can increase response time and server load. A hash-based caching system can improve lookup performance, but multiple product IDs may map to the same hash index, creating collisions.

This project designs a collision-resilient caching system that provides fast product retrieval using a hash table with efficient collision handling and caching techniques. The system is evaluated based on lookup performance, collision handling, cache hit rate, and memory usage.

2. Application of Course / Domain Knowledge

The project applies fundamental data structures and algorithm concepts:

- Hash Tables for efficient product indexing and retrieval.
- Hash Functions for converting product IDs into cache indexes.
- Separate Chaining to handle hash collisions using linked lists.
- LRU Caching to maintain frequently or recently accessed products.
- Linked Lists for collision chains and cache management.
- Algorithm Analysis to compare average and worst-case lookup complexity.

These concepts demonstrate how data structures can solve real-world e-commerce performance problems.

3. Solution Methodology / Design / Implementation

The system follows these steps:

1. Receive a product lookup request.
2. Generate a hash value from the Product ID.
3. Check the corresponding hash table bucket.
4. Search the collision chain for the requested product.
5. If found, return the product as a cache hit.
6. If not found, simulate retrieval from the main storage.
7. Insert the product into the cache.
8. Remove the least recently used product if the cache capacity is full.

Architecture

User Request → Hash Function → Hash Table → Collision Handling
→ Cache Result → Product Response

```

1  #include <stdio.h>
2  #define TABLE_SIZE 10
3
4  int hashTable[TABLE_SIZE];
5
6  int hashFunction(int productID)
7  {
8      return productID % TABLE_SIZE;
9  }
10
11 void insertProduct(int productID)
12 {
13     int index = hashFunction(productID);
14     hashTable[index] = productID;
15     printf("Product %d stored at index %d\n",
16           productID, index);
17 }
18 void searchProduct(int productID)
19 {
20     int index = hashFunction(productID);
21     if (hashTable[index] == productID)
22         printf("Cache Hit: Product Found\n");
23     else
24         printf("Cache Miss: Product Not Found\n");
25 }
26
27 int main()
28 {
29     insertProduct(101);
30     insertProduct(205);
31
32     searchProduct(101);
33     searchProduct(999);
34
35     return 0;
36 }

```

4. Use of Appropriate Modern Tools / Techniques

The project uses the following techniques:

- Hash-based indexing
- Separate chaining for collision handling
- LRU (Least Recently Used) caching strategy
- Modular programming
- Cache hit and cache miss monitoring
- Performance measurement using execution time

Suggested Tools

- **Programming:** C
- **IDE:** Visual Studio Code / Code::Blocks
- **Compiler:** GCC
- **Testing:** Multiple datasets with different product IDs

- **Version Control:** Git/GitHub

No complex modern software is required because the main objective is to demonstrate an efficient data-structure-based system.

5. Results, Testing & Validation

The system is tested using multiple product lookup requests, including product IDs that produce hash collisions.

Testing Parameters

Test	Expected Result
Unique Product ID	Product inserted successfully
Duplicate Product ID	Existing product detected
Hash Collision	Both products stored correctly
Existing Product Lookup	Cache hit
Missing Product Lookup	Cache miss
Full Cache	Least recently used item removed

Performance Metrics

- Cache hit rate
- Cache miss rate
- Number of collisions
- Lookup execution time
- Memory utilization

The validation confirms that collision handling prevents data loss and maintains correct product retrieval.

```
Product 101 stored at index 1
Product 205 stored at index 5
```

```
Cache Hit: Product Found
Cache Miss: Product Not Found
```

6. Analysis, Trade-offs & Justification

The hash-based cache provides an average lookup complexity of **O(1)**, making it suitable for high-throughput applications.

Trade-offs

Approach	Advantage	Limitation
Hash Table	Fast lookup	Collisions may occur
Separate Chaining	Simple and reliable	Additional memory for linked lists
LRU Cache	Keeps useful data	Requires cache maintenance
Small Table	Uses less memory	Higher collision probability
Large Table	Fewer collisions	Increased memory usage

Separate chaining is selected because it can safely store multiple products at the same hash index and allows the cache to continue operating even when collisions occur.

7. Broader Considerations / Professional Responsibility

The system should be designed to handle product data reliably and securely.

Key considerations include:

- Protecting product and user-related data.

- Preventing incorrect product retrieval caused by poor collision handling.
- Avoiding excessive memory consumption.
- Designing scalable solutions for increasing product data.
- Maintaining fair and accurate access to product information.
- Testing the system under high request loads.

Professional responsibility requires developers to validate system correctness before deployment, especially in systems processing large volumes of commercial data.

8. Technical Documentation & Reflection

The project documentation includes:

- Problem definition
- System architecture
- Data structure design
- Hash function
- Collision resolution method
- Cache replacement strategy
- Source code
- Test cases
- Results and analysis

Reflection

This project demonstrates the importance of efficient data structures in real-world software systems. The main learning outcome is understanding that a fast algorithm alone is not sufficient; collision

management, cache capacity, memory usage, and data correctness must also be considered.

The project shows how hash tables combined with collision handling and caching techniques can significantly improve the performance of repeated product lookups in e-commerce applications.